ARTIFICIAL INTELLIGENCE

CO2 – Dry Run Evaluation 2.3: Worked Answers

Prepared from the uploaded assessment. For the graph questions, children are visited left to right. For the 8-puzzle, blank-tile moves are generated in this order: Up, Down, Left, Right.

Question 1 – Breadth-First Search (BFS)

Given graph: A is the root; B and C are its children; D and E are children of B; F and G are children of C.

BFS traversal order: A → B → C → D → E → F → G

Iteration	Queue before processing	Visited node	Queue after adding children
1	[A]	A	[B, C]
2	[B, C]	B	[C, D, E]
3	[C, D, E]	C	[D, E, F, G]
4	[D, E, F, G]	D	[E, F, G]
5	[E, F, G]	E	[F, G]
6	[F, G]	F	[G]
7	[G]	G	[]

Explanation: BFS uses a FIFO queue (First In, First Out). Remove the first node, visit it, and add its unvisited children to the back of the queue.

Question 2 – Depth-First Search (DFS)

Assume left-to-right traversal. DFS explores one branch as deeply as possible before backtracking.

DFS traversal order: A → B → D → E → C → F → G

Step	Node visited	Reason / movement
1	A	Start at the root.
2	B	Move to A's left child.
3	D	Move to B's left child; D is a leaf.
4	E	Backtrack to B and visit its right child.
5	C	Backtrack to A and visit its right child.
6	F	Move to C's left child; F is a leaf.
7	G	Backtrack to C and visit its right child.

Note: DFS commonly uses a stack or recursion, while BFS uses a queue.

Question 3 – 8-Puzzle using BFS

Initial state

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1 3 6
5 0 2
4 7 8
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Goal state

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1 2 3
4 5 6
7 8 0
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The table in the assessment has five rows, but it does not specify a move-generation order. The worked solution below uses Up, Down, Left, Right and shows the shortest path found by BFS.

Shortest path length: 8 moves

Blank moves: Right → Up → Left → Down → Left → Down → Right → Right

State-by-state solution path

Step	Blank move	State (rows separated by /)
Initial	—	1 3 6 / 5 0 2 / 4 7 8
1	Right	1 3 6 / 5 2 0 / 4 7 8
2	Up	1 3 0 / 5 2 6 / 4 7 8
3	Left	1 0 3 / 5 2 6 / 4 7 8
4	Down	1 2 3 / 5 0 6 / 4 7 8
5	Left	1 2 3 / 0 5 6 / 4 7 8
6	Down	1 2 3 / 4 5 6 / 0 7 8
7	Right	1 2 3 / 4 5 6 / 7 0 8
8	Right	1 2 3 / 4 5 6 / 7 8 0

Answers to short questions

1. Which node is expanded in each iteration? In BFS, expand the state at the front of the queue. The exact five rows of the assessment's dry-run table depend on the queue order and whether it asks for the first five expansions or the trace until the goal.

2. How many states are generated in total? The question does not define whether to count all successor states considered or only unique states added to the queue. In the shortest solution path shown above there are 9 states including the initial and goal states; this is the path length count, not the total number of states explored by BFS.

3. Complete solution path: Follow the blank moves and states in the table above, from the initial state to the goal state.

4. Why does BFS find the shortest solution in an unweighted 8-puzzle? BFS explores states in increasing order of depth (number of moves). Since every move has equal cost, the first goal state reached has the minimum number of moves.

Exam tip: Clearly show the queue before and after expansion, mark the state expanded, and keep the same move order throughout.